

We calculate the direct (or “no spillover”) wage increase by moving each missing job under the new minimum wage exactly to the new minimum wage:

$$(3) \quad \% \Delta w_{\text{no spillover}} = \frac{\sum_{k=-4}^{-1} k(\alpha_k - \alpha_{-1k})}{\overline{wb}_{-1}}.$$

The total wage increase of affected workers, $\% \Delta w$, in [equation \(2\)](#) incorporates this direct effect and the add-on effect from wage spillovers. Therefore, the difference between the two measures, $\% \Delta w - \% \Delta w_{\text{no spillover}}$, provides an estimate of the size of the wage spillovers.

Note that our spillover estimates use the frequency distribution of wages, which contrasts with the earlier literature relying on the density of wages (see, e.g., [Card and Krueger 1995](#); [DiNardo, Fortin, and Lemieux 1996](#); [Lee 1999](#); [Autor, Manning, and Smith 2016](#)). As a result, changes in employment—which could create an artificial spillover effect when using the wage density—do not affect our estimates.

We report our estimates of wage spillovers in [Table IV](#), where the columns show estimates of the total wage effect $\% \Delta w$, the “no spillover” wage effect $\% \Delta w_{\text{no spillover}}$, and the spillover share of the total wage increase calculated as $\frac{\% \Delta w - \% \Delta w_{\text{no spillover}}}{\% \Delta w}$. The first row shows the estimated effects for the entire workforce. Column (1) repeats the estimated total wage effect from [Table I](#), column (1), which is 6.8% (std. err. 1.0%). Column (2) shows that in the absence of spillovers, wages would increase by 4.1% (std. err. 0.9%). Column (3) shows that 39.7% (std. err. 11.9%) of the total wage effect is caused by the ripple effect of the minimum wage.

In [Table IV](#) we also report estimates for several subgroups. The share of spillovers in the total wage increase is relatively similar for several key demographic groups, such as those without a high school degree (37.0%), teens (34.7%), those without a college degree (40.2%), and women (35.9%). In most cases, the spillover share is statistically significantly different from 0 at the 5% level. One exception is black or Hispanic individuals, for whom the estimated share of wage spillover is much smaller at 17.9% (std. err. 26.5%), which is less than half of the 39.7% (std. err. 11.9%) spillover share for all workers. Although the difference is not statistically significant, this finding nonetheless suggests that the

TABLE IV
THE SIZE OF THE WAGE SPILLOVERS

	%Δ Affected wage		Spillover share of wage increase
	%Δ <i>w</i>	%Δ <i>w</i> No spillover	$\frac{\% \Delta w - \% \Delta w_{\text{No spillover}}}{\% \Delta w}$
Overall	0.068*** (0.010)	0.041*** (0.009)	0.397*** (0.119)
Less than high school	0.077*** (0.013)	0.048*** (0.009)	0.370*** (0.078)
Teen	0.081*** (0.015)	0.053*** (0.007)	0.347*** (0.059)
High school or less	0.073*** (0.013)	0.043*** (0.011)	0.402*** (0.100)
Women	0.070*** (0.011)	0.045*** (0.010)	0.359*** (0.120)
Black or Hispanic	0.045*** (0.012)	0.037*** (0.010)	0.179 (0.265)
Tradeable	0.058 (0.073)	0.065** (0.028)	−0.114 (1.157)
Nontradeable	0.056*** (0.014)	0.043*** (0.006)	0.237 (0.191)
Incumbent	0.095*** (0.020)	0.055*** (0.011)	0.422** (0.181)
New entrant	0.019 (0.013)	0.023*** (0.006)	−0.178 (0.748)

Notes. The table reports the effects of a minimum wage increase on wages based on the event study analysis (see [equation \(1\)](#)) exploiting 138 state-level minimum wage changes between 1979 and 2016. The table reports the percentage change in affected wages with (column (1)) and without (column (2)) taking spillovers into account for all workers, workers without a high school degree, teens, individuals with high school or less schooling, women, black or Hispanic workers, in tradeable industries, in nontradeable industries, those who were employed one year before the minimum wage increase (incumbents); and those who did not have a job one year before (new entrants). The first column is the estimated change in the affected wages calculated according to [equation \(2\)](#), and the second column assumes no spillovers (see [equation \(3\)](#)). In the last column, the spillover share of the wage effect is calculated by subtracting 1 from the ratio of the estimates in the second to the first column. Robust standard errors in parentheses are clustered by state; significance levels are *0.10, **0.05, ***0.01.

wage gains at the bottom may be more muted for some disadvantaged groups.²⁹

We also find a substantially smaller change in wages due to spillovers in the tradeable sector, though the estimates here are a bit imprecise. This highlights that wage effects are small

29. The smaller spillover for black/Hispanic workers is not due to sectoral or incumbency composition, which are very similar to other workers (results not reported).

in the tradeable sector. The combination of this evidence and the disemployment effects suggest that there may be more unintended consequences of minimum wages when the tradeable sector constitutes a more sizable share of the affected workforce.

We also find a stark difference in the spillover shares of wage increases for incumbents versus new entrants. Incumbents receive a larger total wage increase (9.5%) than the overall workforce (6.8%), but the spillover share for incumbents and all workers is relatively similar (42.2% and 39.7%, respectively). In contrast, the spillover share for entrants is -17.8% , suggesting that essentially all of the wage increase received by new entrants is by the creation of jobs at or very close to the new minimum. Larger spillovers for incumbents relative to entrants can also be seen in [Figure IV](#). Two points should be noted.

First, the stark differences in the size and scope of spillovers for the incumbent and for the new entrants are inconsistent with a simple measurement error process common to both groups. This suggests that spillover effects found are likely to reflect real responses and not measurement error in CPS-based wages, a possibility that is raised by [Autor, Manning, and Smith \(2016\)](#).³⁰

Second, because we find that essentially none of the wage spillovers accrue to workers who were not employed prior to the minimum wage increase, it is unlikely that our estimates of spillovers primarily reflect an increase in the value of the outside options or reservation wages of nonemployed workers (e.g., [Flinn 2006](#)). In contrast, the spillovers may reflect some “optimization friction” that firms face when they set incumbent workers’ wages. [Kleven \(2016\)](#) discusses a range of optimization frictions in the context of bunching at kink points. Moreover, our results are also consistent with [Dube, Giuliano, and Leonard \(2018\)](#) who argue that firms are constrained by relative-pay norms inside the firm.

III.C. Event-Specific Estimates

So far, most of our evidence has come from averaging the effects across all 138 events. In this section, we estimate treatment

30. In [Online Appendix C](#) we implement our approach using administrative data from Washington. In that data we find similar spillover effects which provides additional evidence that the spillovers are not primarily caused by CPS-specific misreporting by survey respondents. In addition, as shown in [Online Appendix Table F.3](#), our wage estimates are similar using a deconvolved distribution that purges the type of measurement error proposed in [Autor, Manning, and Smith \(2016\)](#).

effects for each of the events separately and assess how these impacts vary when we consider minimum wage increases that are more binding.

For this purpose, we create 138 data sets, one for each event h . The data sets include the state of event h and all clean control states for an eight-year panel by event time. Clean control states are those that do not have any nontrivial state minimum wage increases in the eight-year panel around event h ; other states are dropped from data set h . We calculate the event-specific per capita number of jobs in \$1 wage bins relative to the minimum wage for each state-by-year. Then, the regression equation is,

$$(4) \quad Y_{sjth} = \sum_{\tau=-3}^4 \sum_{k=-4}^4 \alpha_{\tau kh} I_{sjth}^{\tau k} + \mu_{sjh} + \rho_{jth} + \Omega_{sjth} + u_{sjth}$$

where j indicates the j th dollar bin relative to the minimum wage. Then, Y_{sjth} is the per capita number of jobs in state s time t , and j th bin relative to the minimum wage in data set h . The calculation of the event-specific change in excess jobs above (Δa_h), change in missing jobs below (Δb_h), and employment change ($\Delta e_h = \Delta a_h + \Delta b_h$) are similar to the ones described in [Section II.B](#). Ω_{sjth} controls for other primary, federal, and small events whose five-year post-treatment periods take place within the data set h . It takes the value of 1 for all post-treatment periods of these events.³¹

[Figure V](#), Panel (A) shows the nonparametric bin-scattered relationship between the event-by-event estimates on missing jobs

31. [Online Appendix](#) Figure D.1 reports event-specific estimates for excess and missing jobs, and employment effects, along with ([Ferman and Pinto, forthcoming](#)) confidence intervals that are appropriate for a single treated unit and heteroskedasticity. Although there is considerable heterogeneity in the bite of the policy, the distribution of employment estimates is consistent with the sharp null of zero effect everywhere: only 5.3% of estimates are statistically significant at the 5% level. In addition, the stacked event-by-event estimates can be also used to estimate the average effect of the minimum wage across events. In [Online Appendix](#) Table D.1 we report estimates using that approach and show that estimates are very similar to our panel regression-based event study. This shows that issues about negative weighting using staggered treatments (e.g., [Abraham and Sun 2018](#)) are unlikely to be driving our results. Finally, the event-by-event estimates in [Online Appendix](#) Figure D.2 confirm that the lack of leading effects and upper-tail employment changes hold event by event, and not just on average: only 5.4% of the events experience statistically significant upper-tail effects at the 5% level, while 7.7% of the events experience statistically significant leading effects. For additional details, see [Online Appendix D](#).